

45 more user needs grouped and ranking (all latent)

It should have flexible electrical applications and built-in protection.

- 11. User can charge high-draw electronics e.g. laptops directly from strip.
- 12. Product should help identify devices that draw power even when they should be off.
- 14. Device must protect connected equipment from damage during low-voltage brownouts or high-voltage spikes.
- 15. Device needs to switch power states without causing electrical arcing to prolong the life of connected appliances.
- 19. Device must maintain power delivery even for devices with very low energy consumption.
- 43. Device needs to filter "dirty electricity" (noise) to protect sensitive audio equipment or computers.
- 40. User needs high-power devices to receive power gradually to prevent breaker popping.

Product should be safe and made with materials that are suitable for its intended use

- 42. User needs a physical way to cut power to all outlets immediately in case of an emergency.
- 44. User needs to know exactly when the surge protection capacity is depleted.
- 16. Device must have internal overheating protection.

Product should be easy to setup and be comfortable to use.

- 20. User must be able to secure power strip to various surfaces.
- 23. Device should help prevent tangling of wires.
- 24. Device should be able to operate without disturbing sleep.
- 25. Visually/auditorily impaired users can still identify and operate the device.
- 29. User needs the device to be well anchored enough to sit on the floor without sliding around when plugging things in.
- 30. Device should allow for some sort of labeling of plugged in devices.
- 33. Device must somehow tell the user what is wrong with it for easy self-diagnosis.
- 39. The user needs to let visitors control the device without giving permanent administrative rights.
- 35. User needs to be able to see currency based on their specific local utility rate.
- 36. User needs to know exactly what changes are being made to the device before agreeing to any externally pushed updates.
- 18. User can access historical usage data to analyze trends over months or years.
- 13. Product should show energy consumption in terms of currency as well as units of energy.
- 22. Allows for devices to be plugged in and always powered, not able to be accidentally turned off.

Product should have a clear and sustainable end of life plan

- 31. User needs clear instructions on how to dismantle and recycle the electronic parts at the end of the product's life

Product should be physically designed well & durable.

- 17. The device must be able to be restored after current overload without needing to purchase replacement parts.
- 27. User should be able to add additional outlets without occupying more wall sockets.
- 21. User needs to prevent small debris or small fingers from entering live electrical contacts.
- 32. Device should be UV resistant.
- 34. Any metal contacts should resist rust if used in humid areas.

Product should easily connect and interface well with external or separate devices

- 5. Product needs to be able to be rapidly and privately controlled without dependence on an external server.
- 7. Product should be just as reliable offline as online if such connectivity exists.

Product should provide advanced automated convenience/safety features

- 1. Product should not require manual input to toggle power.
- 2. Product should be burglar proof.
- 3. Any functionality synchronized with sunrise/sunset should update with changing daylight hours.
- 4. Product needs to react to external data from other apps/services, e.g. weather reports or emails.
- 6. Product allows user to define how it reacts to power outages.
- 8. User should be able to control multiple ports as single functional units.
- 9. The user should be able to define temporary power states as well as predefined schedules, without them altering each other.
- 10. Product should respond to connected appliances no longer drawing power.
- 37. Device should allow for ports to be turned on/off for semi-random periods of time.
- 38. Device allows for one outlet to trigger another in a chain.
- 41. Device needs to recognize specific sounds and respond accordingly, e.g. fire alarms.
- 45. Device should track outages.

New Ranking:

1. Product should be physically designed well and durable.
2. Product should be safe and made with materials that are suitable for its intended use.
3. Product should have a clear and sustainable end of life plan.
3. Product should be easy to setup and be comfortable to use.
4. It should have flexible electrical applications and built in protection.
5. Product should easily connect and interface well with external or separate devices.
6. Product should provide advanced automated convenience/safety features.

Reasoning:

Along with the reasoning described previously, we ranked advanced automated features to be lowest priority because we understand them to not be essential to the original scope of the project. Furthermore, the needs in this meta statement/category was written entirely by AI so it is difficult to verify that they are real explicit/latent needs observed from users.

The meta-need/category about sustainable end of life planning is put below safety and durability because those are essential to the product being made, but they are above the others because we understand the importance of making the product sustainable and as low of a cost to the environment as possible. Ideally we could also conduct a full life cycle analysis if this idea was made into a real product, to ensure that we follow this need.